Table R1: **Table 2 of the submission with the refined rule ($\delta = 0.3$) as the default.** Counterfactual Video Foley Generation on 11,890 VGGSound-Counterfactual triplets. Panel (b) uses comparable NFE, with doubled steps for Default CFG and negative-source CFG and the published transition for CounterFlow. Bold and underline mark the best and second-best values *across all rows in (a)* and *within each backbone in (b)*. Blue marks the new values of the refined rule: the **TAP-Foley** rows use the margin rule with $\delta = 0.3$. The rows marked "$\alpha = 1.2$, submitted" are the submitted scaled projection, kept for reference with their submitted values (black) and not ranked. Baseline rows, including their bold and underline marks, are exactly as submitted. †FAD is not ranked.

Method		Sound Identity Replacement					Temporal Alignment	Distribution /Quality		
		CLAP-T ↑	Clf. acc. ↑	FLAM-T ↑	ΔFLAM ↑	(+) Ratio ↑	DeSync ↓	IS ↑	PQ ↑	FAD†
(a) Comparison with existing VT2A systems										
Ours										
TAP-Foley	MMAudio	**0.299**	**0.901**	**0.960**	0.425	**0.916**	0.455	**21.16**	6.21	22.8
	Omni2Sound	0.242	0.833	0.891	0.301	0.824	0.501	19.98	6.30	24.7
TAP-Foley (α=1.2, submitted)	MMAudio	0.296	0.885	0.959	0.384	0.900	0.427	20.64	6.21	21.6
	Omni2Sound	0.241	0.829	0.887	0.299	0.821	0.510	19.83	6.31	24.5
Baseline										
ControlFoley		0.229	0.601	0.864	0.089	0.610	**0.316**	17.10	6.13	14.6
CAFA		0.243	0.786	0.942	0.194	0.785	0.685	12.21	6.36	22.3
FoleyCrafter		0.154	0.514	0.746	-0.004	0.478	1.244	10.74	6.06	30.4
ReWaS		0.005	0.443	0.344	-0.005	0.504	1.088	5.33	5.44	58.3
AudioX		0.187	0.475	0.840	-0.034	0.475	1.136	16.27	**6.53**	18.1
Baseline (with negative-source CFG)										
ControlFoley		0.255	**0.893**	0.910	**0.456**	**0.916**	0.754	15.66	5.89	23.6
CAFA		0.200	0.873	0.875	0.335	0.860	0.715	10.36	6.25	24.5
FoleyCrafter		0.105	0.732	0.662	0.281	0.742	1.264	8.08	6.02	40.8
ReWaS		-0.045	0.586	0.190	0.050	0.575	1.168	4.73	5.52	67.7
AudioX		0.217	0.845	0.873	0.329	0.853	1.233	14.79	6.19	25.2
(b) Comparison of inference-time guidance methods										
MMAudio										
TAP-Foley (ours)		**0.299**	0.901	**0.960**	0.425	0.916	0.455	**21.16**	**6.21**	22.8
TAP-Foley (α=1.2, submitted)		0.296	0.885	0.959	0.384	0.900	0.427	20.64	6.21	21.6
CounterFlow		0.292	**0.916**	0.949	**0.475**	**0.927**	0.649	19.75	6.16	23.7
Default CFG		0.162	0.350	0.727	-0.145	0.335	**0.323**	16.39	**6.21**	10.8
Negative-source CFG		0.212	0.758	0.824	0.280	0.778	0.703	12.93	5.93	19.2
Omni2Sound										
TAP-Foley (ours)		**0.242**	**0.833**	**0.891**	**0.301**	**0.824**	0.501	**19.98**	**6.30**	24.7
TAP-Foley (α=1.2, submitted)		0.241	0.829	0.887	0.299	0.821	0.510	19.83	6.31	24.5
CounterFlow		0.213	0.765	0.862	0.226	0.740	0.576	15.00	6.06	23.7
Default CFG		0.110	0.223	0.620	-0.267	0.195	**0.408**	18.58	6.27	14.2
Negative-source CFG		0.132	0.453	0.714	-0.029	0.438	0.642	11.35	6.05	27.1

Table R2: **Table 7 of the submission with the refined rule** ($\delta = 0.3$)**.** Onset-level evaluation on 1,280 Greatest Hits counterfactual triplets. Onset scores count only detected onsets with PE-A-Frame evidence for the target identity, so they measure identity and timing jointly (target-filtered onset metrics, appendix of the submission). Recall, precision, and F1 are the standard event-level scores over reference onsets, from a maximum one-to-one matching within a 0.3-s tolerance. Panel (a) covers the two conflict-aware systems. Panel (b) uses comparable NFE and the published transition for CounterFlow. Bold and underline mark the best and second-best values *across all rows in (a)* and *within each backbone in (b)*. Blue marks the new values of the refined rule: the **TAP-Foley** rows use the margin rule with $\delta = 0.3$. The rows marked "$\alpha = 1.2$, submitted" are the submitted scaled projection, kept for reference with their submitted values (black) and not ranked. Baseline rows, including their bold and underline marks, are exactly as submitted.

Method		Sound Identity Replacement		Onset Analysis		
		CLAP-T ↑	Clf. acc. ↑	Onset recall ↑	Onset precision ↑	Onset F1 ↑
(a) Comparison with existing VT2A systems						
Ours						
TAP-Foley	MMAudio	**0.347**	**0.997**	**0.864**	0.802	**0.797**
	Omni2Sound	0.303	0.952	0.809	0.775	0.757
TAP-Foley (α=1.2, submitted)	MMAudio	0.343	0.991	0.851	0.818	0.801
	Omni2Sound	0.297	0.948	0.803	0.780	0.758
Baseline						
ControlFoley		0.301	0.881	0.771	**0.833**	0.769
CAFA		0.305	0.956	0.775	0.802	0.756
Baseline (with negative-source CFG)						
ControlFoley		0.305	0.981	0.754	0.632	0.616
CAFA		0.276	0.976	0.767	0.722	0.704
(b) Comparison of inference-time guidance methods						
MMAudio						
TAP-Foley		**0.347**	**0.997**	**0.864**	0.802	**0.797**
TAP-Foley (α=1.2, submitted)		0.343	0.991	0.851	0.818	0.801
CounterFlow		0.305	0.981	0.705	0.673	0.631
Default CFG		0.295	0.877	0.746	**0.830**	0.751
Negative-source CFG		0.324	0.984	0.798	0.724	0.716
Omni2Sound						
TAP-Foley		**0.303**	**0.952**	**0.809**	**0.775**	**0.757**
TAP-Foley (α=1.2, submitted)		0.297	0.948	0.803	0.780	0.758
CounterFlow		0.249	0.913	0.713	0.632	0.612
Default CFG		0.174	0.541	0.495	0.574	0.506
Negative-source CFG		0.184	0.823	0.603	0.639	0.581

Table R3: **Family-stratified means on the full VGGSound-Counterfactual benchmark (refined rule, $\delta = 0.3$, matched NFE).** Mean ΔFLAM and DeSync of Default CFG, CounterFlow and **TAP-Foley** over the pairs (video, target prompt) of each stratum; Default CFG and CounterFlow use the matched-NFE setting of Table 2(b), and the first row of each block is the full benchmark. Strata: the family of the source category, the family of the target prompt (the seven sound families of App. C.1, incl. the catch-all Other), and the family relation (within-family: source and target in the same family). In every stratum, **TAP-Foley** replaces more than Default CFG (higher ΔFLAM) and has lower DeSync than CounterFlow, as on the full benchmark; † marks the few **TAP-Foley** values whose paired difference to that comparator is not significant (95% percentile CI from a cluster bootstrap over the source videos of the stratum, 10,000 draws, includes 0).

Stratum	Pairs	ΔFLAM ↑			DeSync ↓		
		Default CFG	CounterFlow	**TAP-Foley**	Default CFG	CounterFlow	**TAP-Foley**
MMAudio							
Full benchmark	11,890	-0.145	0.475	0.425	0.323	0.649	0.455
Source family							
Animals	1,890	-0.067	0.540	0.483	0.352	0.733	0.519
Human voice/vocal	880	-0.289	0.513	0.402	0.117	0.426	0.234
Instruments	4,490	-0.161	0.367	0.323	0.263	0.533	0.364
Other	3,340	-0.118	0.574	0.529	0.366	0.716	0.517
Sports	430	-0.143	0.330	0.322	0.328	0.695	0.441
Vehicles	650	-0.202	0.582	0.567	0.650	1.037	0.852
Water/weather	210	-0.171	0.402	0.311	0.481	0.936	0.582
Target family							
Animals	2,229	-0.132	0.557	0.477	0.291	0.661	0.458
Human voice/vocal	554	-0.324	0.518	0.456	0.154	0.401	0.258
Instruments	4,158	-0.108	0.358	0.313	0.233	0.509	0.332
Other	3,319	-0.147	0.522	0.492	0.428	0.786	0.570
Sports	707	-0.227	0.403	0.353	0.333	0.704	0.502
Vehicles	617	-0.206	0.683	0.633	0.512	0.875	0.678
Water/weather	306	-0.093	0.612	0.549	0.519	0.831	0.672
Family relation							
Within-family	8,060	-0.091	0.410	0.366	0.293	0.590	0.401
Cross-family	3,830	-0.260	0.610	0.550	0.384	0.771	0.568
Omni2Sound							
Full benchmark	11,890	-0.267	0.226	0.301	0.407	0.576	0.501
Source family							
Animals	1,890	-0.264	0.306	0.387	0.476	0.695	0.593
Human voice/vocal	880	-0.401	0.292	0.428	0.226	0.365	0.319
Instruments	4,490	-0.264	0.151	0.233	0.313	0.438	0.406
Other	3,340	-0.232	0.269	0.326	0.463	0.663	0.556
Sports	430	-0.135	0.091	0.162	0.409	0.509	0.482†
Vehicles	650	-0.349	0.306	0.336	0.731	0.905	0.754
Water/weather	210	-0.393	0.186	0.218	0.680	1.079	0.874
Target family							
Animals	2,229	-0.254	0.360	0.432	0.406	0.632	0.549
Human voice/vocal	554	-0.406	0.286	0.432	0.263	0.370	0.319
Instruments	4,158	-0.216	0.147	0.221	0.295	0.409	0.371
Other	3,319	-0.328	0.198	0.261	0.522	0.727	0.610
Sports	707	-0.247	0.182	0.257	0.389	0.486	0.472†
Vehicles	617	-0.295	0.287	0.402	0.586	0.790	0.671
Water/weather	306	-0.136	0.498	0.517	0.652	0.953	0.806
Family relation							
Within-family	8,060	-0.225	0.175	0.244	0.377	0.530	0.456
Cross-family	3,830	-0.356	0.334	0.421	0.472	0.674	0.598